Classical Field Theory — Exercise Sheet 6

Causal kernels and frequency boundary values

Hints and answers

Use $\Box = \partial_t^2 - \Delta$ and Lecture 6's normalized kernel $G_{\mathrm{ret}} = \delta(t-|x|)/(4\pi|x|)$ in three spatial dimensions. Its meaning at the vertex is its test pairing, not a product of distributions. All distribution pairings are bilinear. Write H for the Heaviside function; its value at zero does not affect the distributions below.

Exercise 6.1 (An entire radial source response). Let $q \in C_c^\infty(\mathbb{R})$, and let h be a smooth even compactly supported function such that $x \mapsto h(|x|)$ is smooth on $\mathbb{R}^3$. Take $f(t,x) = q(t)h(|x|)$ and $u = G_{\mathrm{ret}} * f$.

(a) Perform the angular integration to obtain a one-dimensional radial integral for $u(t,r)$, $r = |x| > 0$, with an inner integral over travel distance. Find its limit at $r = 0$ and explain why no singularity remains there.

(b) Set $v(t,r) = ru(t,r)$ for $r \geq 0$ and extend it oddly to the real line. Reduce the equation for v to a forced one-dimensional wave and derive its causal double-integral formula. Check its agreement with part (a).

(c) Let $a < b$ and $R > 0$. If $\operatorname{supp} q \subset [a,b]$ and $\operatorname{supp} h \subset [-R,R]$, find the possible response interval at an observer $r > R$. If $q > 0$ on (a,b) and $h > 0$ for $|s| < R$, prove that the response is positive throughout the interior of that interval.

Hint. At fixed source radius s, use the travel distance $\ell = (r^2 + s^2 - 2rs\cos\theta)^{1/2}$ as angular variable. The one-dimensional forcing is the odd function $sh(|s|)q(t)$.

Answer. $u(t,r) = \frac{1}{2r}\int_0^\infty sh(s)\int_{|r-s|}^{r+s} q(t-\ell)\mathrm{d}\ell\mathrm{d}s$ and $u(t,0) = \int_0^\infty sh(s)q(t-s)\mathrm{d}s$. The odd field satisfies $v_{tt} - v_{rr} = rh(|r|)q(t)$. For $r > R$ the response is supported in $[a+r-R, b+r+R]$.

Exercise 6.2 (Which side of the frequency poles?). The spacetime Fourier convention is

$$\widetilde{F}(\nu,k) = (2\pi)^{-2}\int e^{i\nu t - ik\cdot x}F(t,x)\mathrm{d}t\mathrm{d}^3x.$$

For a tempered distribution it is defined by transposition on Schwartz tests, with this same kernel. Fourier transformation preserves Schwartz space and its topology, so this definition preserves convergence of tempered distributions. The Schwartz Fourier automorphism is the one from *Analysis* I used in Lecture 4; transposition here specifies its distributional extension. Let $G_\varepsilon = e^{-\varepsilon t}G_{\mathrm{ret}}$, $\varepsilon > 0$; the product is understood on the future cone, where the factor is bounded.

(a) Compute its spatial transform, then its full spacetime transform. Prove convergence as $\varepsilon \downarrow 0$ on Schwartz tests and express $\widetilde{G}_{\mathrm{ret}}$ as a frequency boundary value. Check the Fourier coefficient of the equation $\Box G_{\mathrm{ret}} = \delta^4$.

(b) For fixed $\kappa = |k| > 0$, derive the principal-value and delta terms of that boundary value. Here $\langle \mathrm{pv}(1/s), \zeta\rangle = \lim_{a\downarrow 0}\int_{|s|>a}\zeta(s)/s\,\mathrm{d}s$. Derive the identity for $(s \pm i0)^{-1}$ by separating real and imaginary parts. Compare the advanced and half-retarded plus half-advanced kernels. State why the fixed-κ decomposition cannot simply be evaluated at $\kappa = 0$.

Hint. The spatial transform of the spherical shell uses $\int_{S^2} e^{-it\kappa\cdot\theta}\mathrm{d}\Omega = 4\pi\,\sin(t\kappa)/(t\kappa)$. Then integrate $e^{-(\varepsilon-i\nu)t}\sin(t\kappa)/\kappa$. For the pole identity, $\varepsilon/(s^2+\varepsilon^2)$ is an approximate delta with integral π.

Answer. $\widetilde{G}_\varepsilon = (2\pi)^{-2}[|k|^2-(\nu+i\varepsilon)^2]^{-1}$. At $\kappa>0$ the boundary value inside the brackets is $\mathrm{pv}(\kappa^2-\nu^2)^{-1}+\frac{i\pi}{2\kappa}[\delta(\nu-\kappa)-\delta(\nu+\kappa)]$. Advanced reverses the imaginary term; their half-sum retains the principal value.

Exercise 6.3 (Descending the spatial dimension). Starting with the three-dimensional retarded pairing, integrate out one auxiliary spatial coordinate to construct a kernel G_2 on $\mathbb{R}^{1+2}$, and two auxiliary coordinates to construct G_1 on $\mathbb{R}^{1+1}$. Make the construction legitimate on compact smooth tests: a test independent of the auxiliary coordinates is not itself compactly supported. Derive explicit locally integrable formulas, prove $(\partial_t^2-\Delta_d)G_d=\delta^{1+d}$ for $d=1,2$, and identify their support and their relation to Sheet 5's interior tails.

Hint. Lift a compact test with an auxiliary cutoff equal to one on a ball larger than its time support. On the cone, the auxiliary radius is bounded by that time support. For two removed coordinates use polar coordinates and then the full travel distance as integration variable.

Answer. $G_2(t,x)=H(t-|x|)/(2\pi\sqrt{t^2-|x|^2})$ inside the future cone, and $G_1(t,x)=H(t-|x|)/2$. Both are retarded fundamental solutions, with support the whole closed future cone.